

Captivating the Senses: Crafting a Multisensory Virtual Experience for Enhanced Realism

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ABSTRACT

This paper explores the challenges and opportunities in achieving a high level of realism and presence in Virtual Reality (VR) experiences. Realism and presence, which refer to the level of authenticity of the virtual environment and the user's subjective perception of being in it, respectively, are fundamental components of engaging and satisfactory VR experiences. Despite advancements in VR technology, users can often distinguish between real and virtual experiences, thereby limiting the potential of VR in various fields. This paper reviews various challenges inherent in achieving indistinguishable realism in VR, encompassing visual, auditory, and tactile components. These insights are critical in guiding the continued development of more immersive and realistic VR experiences.

CCS CONCEPTS

• **Computing methodologies** → *Mixed / augmented reality; Virtual reality*; • **Human-centered computing** → *Mixed / augmented reality; Virtual reality*.

KEYWORDS

Virtual Reality; Mixed Reality; Augmented Reality; Realism; Presence; Spatial Audio; Haptics.

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1 INTRODUCTION

Virtual Reality (VR) and Augmented Reality (AR) hold the potential to redefine our world, possibly surpassing the impact of personal computing. The crux of realizing this potential lies in achieving indistinguishably realistic visual experiences, which contribute significantly to the sense of presence within VR environments. These two essential components, realism and presence, greatly influence user engagement and satisfaction within a VR experience. Realism refers to the extent to which a virtual environment emulates the real world, incorporating visual, auditory, and haptic stimuli to

create a convincing experience [2]. Conversely, presence is a user's subjective perception of being in the virtual environment, closely tied to their sense of immersion and illusion [44].

While current VR systems successfully create a strong sense of presence, they have yet to reach a level where users cannot distinguish between real and virtual experiences. The importance of realism in VR cannot be overstated, as the relationship between realism and presence, though complex, often significantly affects the overall VR experience. Research has shown that increasing the level of realism in VR experiences can sometimes lead to an increase in presence [6, 16, 27, 44]. However, this relationship is context-dependent, suggesting that enhanced visual and auditory cues, the addition of tactile cues, gesture input support, or other sensory inputs may improve realism without necessarily affecting presence [14, 15, 28].

Achieving higher levels of realism will unlock a myriad of VR capabilities, encompassing virtual workspaces that make remote work as productive as, if not more than, working in a physical office space. Furthermore, realistic VR experiences enable authentic virtual social interactions, virtual tourism, and a wide range of applications that mirror our real-world experiences. To realize this, researchers must overcome several challenges, not only in passing the visual Turing test [19]—creating a VR display that is virtually indistinguishable from reality—but also addressing a multitude of other sensory and interaction aspects that contribute to the overall sense of presence.

This paper aims to provide a comprehensive understanding of the multifaceted perceptual approach necessary for overcoming these challenges inherent in achieving a fully immersive and realistic VR experience. Such insights can contribute to the continued advancement of VR technology and its potential applications in various fields.

2 VISUAL CHALLENGES

The process of vision begins with light entering the eye, which is captured by the cornea and focused onto the retina at the back of the eye. The retina is lined with photoreceptor cells, comprising rods and cones, that are sensitive to different aspects of light. These photoreceptor cells convert the light into electrochemical signals, which are then transmitted via the optic nerve to the brain, where they are processed and interpreted as visual images.

As such, the intricacies of the human visual system present unique challenges for VR experiences. As we delve into the visual challenges of achieving true presence, it becomes evident that resolution is merely the tip of the iceberg. While resolution is a notable limitation [26], there are numerous other factors at play, such as vergence-accommodation conflict, chromatic aberration,

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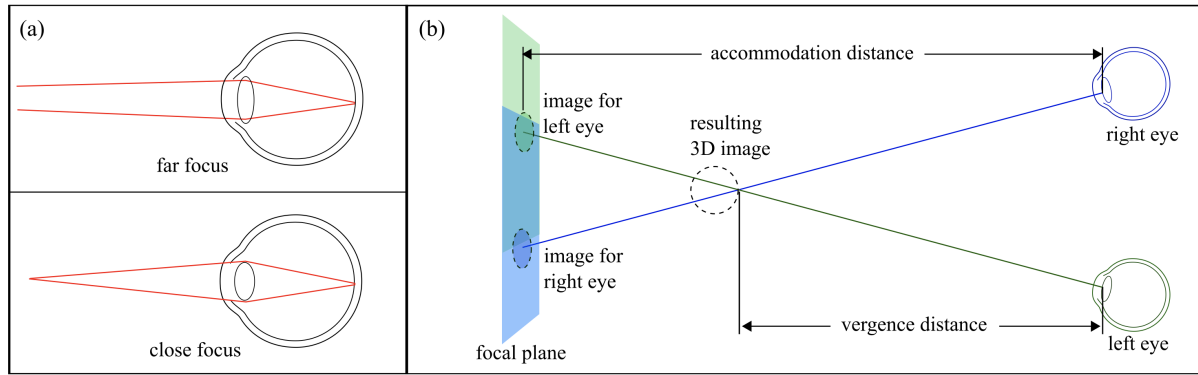


Figure 1: (A) Depiction of eye accommodation for different distances [17]. The human eye lens adjusts its curvature to focus light from objects at varying distances; distant objects need less lens curvature, while nearby objects require more. (B) Illustration of the Vergence-Accommodation Conflict (VAC) in virtual displays, showing a fixed-distance virtual focal plane with objects located in front or behind. Disparity cues lead the eyes to verge at one distance, while retinal blur from the virtual plane prompts accommodation at another, creating a conflict due to the difference in vergence and accommodation distances.

ocular parallax, and pupil swim, which do not exist in traditional 2D displays [20]. Field of view (FoV) has also shown to affect immersion [8], but only affects presence when the user feels agency [13]. Nevertheless, a human eye's FoV is 220° and most commercial VR headsets in the market today are limited to 110° , indicating that further advancements are needed to bridge this gap.

The wider fields of view inherent in VR headsets further exacerbate the resolution issue, necessitating a staggering 13,000 pixels horizontally to achieve 20/20 vision across the entire field [34]. In addition to the sheer number of pixels required, the quality of these pixels must also be enhanced, as current VR headsets lack the brightness and contrast of laptops and other conventional displays [46]. Lens distortion is another major obstacle, as it reduces realism and necessitates complex software-based correction that adapts to the user's eye movement. Furthermore, the ability to focus at varying distances is a critical aspect of visual fidelity.

To surmount these challenges and achieve a visually indistinguishable VR experience, a new technological stack must be developed that will encompass varifocal technology for accurate depth of focus, resolution that surpasses human vision, distortion correction to mitigate optical aberrations, and high dynamic range to expand the range of color, brightness, and contrast experienced. By addressing these visual challenges, we move closer to passing the visual Turing test. In this paper, I will focus on challenges specific to VR, the vergence-accommodation conflict and optical aberrations.

2.1 Vergence-accommodation Conflict

In the real world, our eyes naturally adjust the shape of their lenses to focus on objects at different distances, ensuring that the light entering our eyes is accurately imaged. However, current VR headsets employ fixed-focus optics, typically set at a distance of 5 to 6.5 feet (1.5 to 2 meters). This means that regardless of where we look in a virtual environment, the light effectively originates from the same distance, an unfamiliar phenomenon for our visual systems. This phenomenon is best illustrated in Figure 1, borrowed from [17].

This disparity between the simulated distance of a virtual 3D object and the fixed focusing distance in today's headsets can result in a vergence-accommodation conflict (VAC). VAC may cause temporary fatigue and blurry vision, detracting from the overall VR experience. Although, it can be tackled using one of three strategies [17]: (a) employing discretized addressable focal planes in multi-focal-plane displays, (b) continuously modulating the focal plane for desired accommodation response, or (c) presenting uniform focus in accommodation-free displays. If the user's gaze point is known, virtual content depth can be ascertained, allowing tunable optics to continuously adjust the focal plane accordingly. Consequently, certain tunable lens prototypes have been designed to function in either variable-focal-plane mode or multi-focal-plane mode [23, 24].

The variable-focal-plane or "varifocal" involves dynamically adjusting the focal depth in VR. This method entails moving the lenses in accordance with the viewer's gaze, enabling them to focus on objects at different distances. By incorporating varifocal technology, specifically electronic varifocal [22], we can significantly enhance the comfort and immersion of VR experiences.

2.2 Optical Aberrations

In VR environments, the quality of the image plays a crucial role in maintaining the illusion of reality. Unfortunately, due to technical limitations, no VR lens can be completely free from optical aberrations. Current VR headsets use distortion correction software to minimize these imperfections; however, this software is not without its flaws [3, 45]. One notable issue is the phenomenon known as pupil swim, where the distortion of the virtual image is dynamic, changing depending on where the user is looking [12]. This movement can cause the environment to appear less real, as everything shifts slightly when the user's eye moves, ultimately affecting the overall immersion experience. Figure 2 best illustrates pupil swim during vestibulo-ocular reflex, a type of eye movement where gaze is fixated on a virtual object while the head is moving. Past work has addressed pupil swim by dynamically correcting aberrations by incorporating the user's real-time gaze data [33]. Furthermore,

Guan et al. [10] created a simulator to simulate dynamic optical distortions using a 3D TV, allowing for rapid prototyping of distortion correction software with respect to the lens' optical characteristics. In short, addressing optical aberrations and pupil swim in VR environments dynamically is key for enhancing the overall immersion experience.

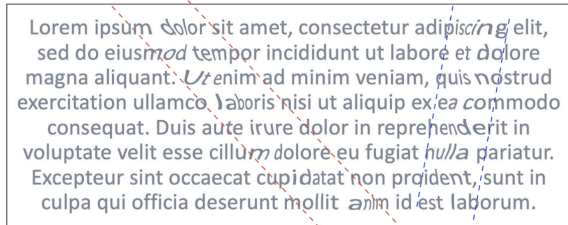


Figure 2: Illustration of the pupil swim phenomenon [12] during vestibulo-ocular reflex. an eye movement where gaze is fixated on an object while the head moves. The regions between the dotted lines depict the distortions.

The brain is continuously interpreting and integrating sensory information from the eyes, as well as other senses, to construct a coherent understanding of the environment. The visual challenges we discussed, such as high dynamic range, VAC, and optical distortions, attempt to simulate this sensory input effectively enough to create a believable experience. When successful, the brain interprets the virtual environment as if it were real, which is known as the suspension of disbelief [7]. By tapping into these perceptual processes, VR can induce a convincing sense of presence and immersion.

3 AUDITORY CHALLENGES

The role of audio in enhancing presence and immersion in virtual environments is crucial, yet it often faces unique challenges in terms of design, implementation, and user experience. By understanding these challenges and their implications, we can better optimize the auditory experience in virtual environments, ultimately leading to a more immersive and realistic experience for users.

3.1 Spatial Audio and Localization

Spatial audio and localization play a crucial role in enhancing auditory immersion as they contribute to the accurate placement and perception of sound sources within the virtual environment. Spatial audio rendering techniques attempt to simulate the way sounds interact with the listener's ears and head, providing a convincing impression of the sound's origin and direction.

There are two main approaches to spatial audio: sound field-related delivery methods and head-related audio rendering methods [31]. Sound field-related methods use multi-channel loudspeaker systems to create a natural-sounding spatial environment, with the speaker placement and room acoustics being essential for proper sound localization [42]. Head-related audio rendering methods, on the other hand, use headphones to control the sound arriving at each ear, enabling precise manipulation of binaural cues and eliminating environmental factors such as echoes and reverberation [21]. Both approaches aim to provide a sense of auditory spatial awareness and

localization, helping users navigate and interact with the virtual environment more effectively. By incorporating spatial audio and localization cues, VR can create a more immersive and realistic auditory experience.

3.2 Immersive soundscapes and Environmental design

Immersive soundscapes and environmental design are also essential for auditory immersion. Soundscapes encompass all auditory elements, such as ambient sounds, sound effects, and music, while environmental design focuses on the spatial organization of sound sources [30]. Designing immersive soundscapes involves carefully selecting and balancing auditory elements to convey the intended mood and atmosphere. For example, Project Triton [35, 36, 38] simulates real wave physics in 3D environments to model immersive sound effects such as occlusion and reverberation. Ambient sounds provide a sense of realism, while sound effects enhance the user's sense of agency and involvement. Strategic placement of sound sources can guide users' attention and facilitate navigation within the virtual world [32].

These audio techniques fool the ear and brain into perceiving a real space by mimicking how sounds behave in real-world environments. By recreating spatial cues, such as sound source direction, distance, and reflections, the brain is tricked into interpreting the virtual soundscape as a convincing representation of a physical space, thus enhancing the user's sense of presence and immersion.

4 HAPTIC CHALLENGES

Real-world perception is typically multimodal as our perceptual apparatus has evolved to make sense of sensations that originate naturally and coherently from a single event, even if received through multiple modalities [29]. In VR, visual, auditory, and haptic feedback must be actively coordinated to create a cohesive experience that mimics this natural coherence.

To understand the complexity of designing effective haptic feedback for virtual experiences, it is essential to examine the underlying mechanisms of tactile perception. The skin is equipped with numerous tactile receptors, which are specialized end organs of the peripheral nervous system. These receptors facilitate the conversion of mechanical, thermal, noxious, itching, or chemical stimuli into neural information. Specific tactile neurons then transform these mechanical signals into electrochemical signals, generating action potentials or spikes that are sent to the brain.

Hence, one of the primary challenges [5] in developing haptic technologies is the creation of general-purpose haptic displays that mimic the sensations experienced when touching real objects. Achieving a high-fidelity haptic display capable of reproducing authentic touch experiences necessitates a method for applying distributed mechanical stimuli, such as forces or displacements, to the skin. This method should allow for exceptionally fine control over the stimuli's temporal and spatial distribution, as well as a wide dynamic range and surface area coverage on the moving body.

Various haptic technologies have been explored to address presence and immersion in VR experiences, as depicted in Figure 3. Vibrotactile feedback devices, such as gloves equipped with spatial tracking and vibration actuators at the fingertips, are one such

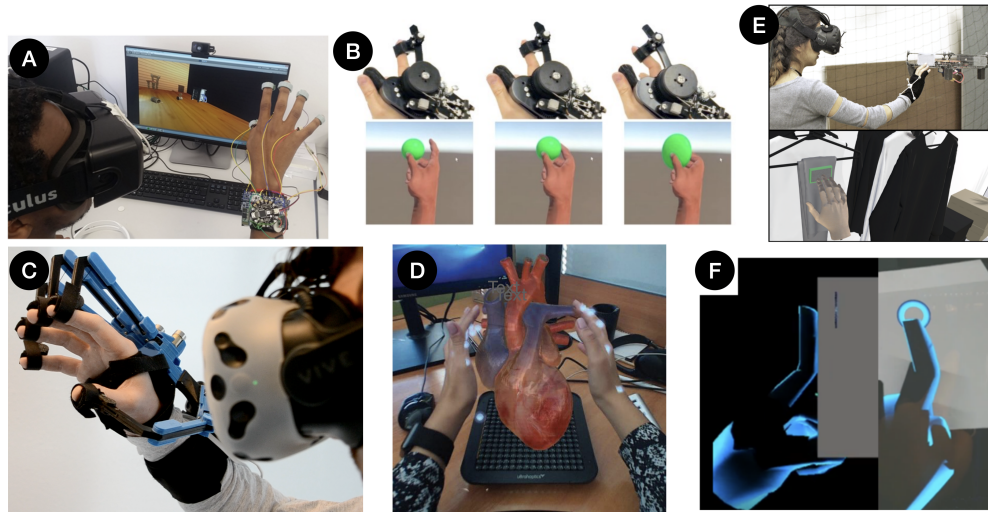


Figure 3: (A) A wearable vibrotactile haptic device with actuators at the finger tips [25]. (B) CapstanCrunch, a forced feedback haptic controller that renders touch and grasp haptic sensations [43]. (C) The Senseglove DK1, a commercial haptic glove that provides forced feedback and tactile stimulation for VR interactions [41]. (D) An implementation of ultrasound haptic feedback for an AR medical visualization [39]. (E) Haptic interactions using a quadcopter - A user is touching fabrics attached to the quad for texture rendering [1]. (F) Visual haptic feedback depicting the user's virtual hand bending against the virtual content during hand-tracked target selection [37].

technology. Force feedback exoskeletons are another approach to providing haptic feedback, like the Senseglove [41] depicted in Figure 3(C), provide haptic feedback through mechanical resistance. Ultrasound haptics is an approach which uses focused ultrasonic waves to create tactile sensations on the user's skin without physical contact [39]. These devices aim to improve tactile sensitivity and provide feedback in various application areas.

Biocca et al. [4]'s work on inter-sensory integration revealed that the illusion of haptic feedback can be induced through visual cues, while audio cues were found to be ineffective in achieving this effect. Accordingly, researchers [37] have implemented visual haptics to provide a sense of touch for mid-air selections. In a recent study by [18], which compared vibrotactile, force feedback, and visual-only conditions, the findings were inconclusive with regard to presence. Nonetheless, some evidence suggested that force feedback led to improved performance in throwing and stacking tasks when compared to vibrotactile and visual-only conditions.

Despite the advancements in haptic technologies, several challenges remain in integrating them into immersive VR experiences. Firstly, achieving accurate spatial alignment of real and virtual objects is crucial for maintaining presence [9]. Inconsistencies and incongruences between visual and haptic information can negatively impact the user's sense of presence. Secondly, the quality of haptic feedback can significantly affect presence, immersion, and even task performance [11, 18, 40].

Haptic technologies play a pivotal role in enriching presence and immersion, although, several specific challenges must be addressed, including accurately aligning real and virtual objects, refining the quality of haptic feedback, and achieving precise control over the temporal and spatial distribution of mechanical stimuli to meet the expectations of our perceptual systems.

5 CONCLUSION AND FUTURE RESEARCH DIRECTIONS

In conclusion, VR has made significant strides in recent years, but several challenges remain in the quest for truly immersive experiences that seamlessly blend with users' natural perception. This paper has explored the issues faced in visual, auditory, and haptic domains and their implications for presence and immersion in virtual environments.

Future research directions should focus on refining and expanding the techniques discussed in this paper to achieve more robust, natural, and seamless immersive experiences. Incorporating eye-tracking technology into VR head-mounted displays is fundamental for addressing visual challenges like VAC and optical aberrations, as it offers invaluable insights into the user's gaze point and focus. The development of spatial audio and localization can benefit from advancements in real-time audio processing and rendering, as well as machine learning algorithms that can accurately model the complexities of sound propagation in virtual environments. Additionally, exploring novel methods of creating immersive soundscapes and environmental design will contribute to enhancing the auditory experience in VR.

In the haptic domain, research should concentrate on the development of high-fidelity, general-purpose haptic displays that can closely mimic real-world tactile sensations. This may involve exploring new materials, actuators, and design principles for wearable haptic devices, as well as investigating methods for improving the spatial and temporal control of mechanical stimuli applied to the skin.

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